

Adolescent Mental Health at Scale: Measurement, Drivers, and System Insight

Established a scalable framework for assessing student mental health and uncovering key behavioural and environmental drivers of wellbeing. By linking risk patterns to academic, family, and lifestyle factors, it enabled more targeted and preventative responses within schools.

Research Context

- **Setting:** Secondary school mental health screening in China, covering students aged 10–17 across key developmental stages
- **Structural tension:** Schools are required to demonstrate measurable safeguarding outcomes, while also addressing student wellbeing in a meaningful, context-sensitive way
- **Limitations of existing approaches:**
 - Screening often functions as one-off detection, with limited insight into underlying causes
 - Psychological risk is typically treated as individual-level, without sufficient attention to environmental and behavioural drivers
- **Core need:**
A scalable approach that not only identifies at-risk students, but also captures the drivers of wellbeing, enabling more informed and preventative responses at the school level

Research Objective

- **Screening (identification):**
 - Systematically identify students at elevated psychological risk to support timely follow-up and prioritisation of limited school resources
- **Measurement (baseline profiling):**
 - Establish a structured, population-level baseline of student mental health across multiple dimensions, enabling comparison across age, gender, and cohort groups
- **Driver analysis (explanatory layer):**
 - Examine how academic pressure, family context, and lifestyle factors relate to mental health outcomes, moving beyond detection toward understanding underlying contributors
- **System insight (application):**
 - Generate evidence that can inform both individual-level intervention and broader school-level decision-making, bridging screening outputs with actionable response strategies

Instrument Design

- **Core scales:**
 - Standardised mental health and stressor measures to ensure reliability and comparability across participants
- **Dimensions:**
 - Multi-dimensional coverage including anxiety, interpersonal sensitivity, somatic symptoms, and behavioural tendencies
- **Contextual variables:**
 - Additional items capturing academic pressure, family environment, and lifestyle factors to enable analysis of potential drivers of wellbeing

Data Collection Approach

- **Scale:**
 - ~1,488 students (ages 10–17) separated into two year groups
- **Delivery:**

- Browser-based system (computer-first), with mobile as supplementary access to ensure coverage and accessibility
- **Data quality control:**
 - Removal of invalid responses (~10%) using validity checks to improve reliability and reduce response bias

Analysis & Methods

- **Tools:** R used for structured statistical analysis
- **Approach:** Combined descriptive stats, group comparisons, and regression to detect patterns
- **Focus:**
 - Identifying population-level trends and differences across age, gender, and context,
 - exploring potential contributors to overall mental health through relationships between stressors and outcomes

Key Findings (high-level)

- **Risk distribution:** Identified both at-risk and borderline groups requiring attention
- **Group differences:** Older students showed consistently higher psychological risk levels
- **Key drivers:** Academic pressure, internal psychological strain, and family environment emerged as dominant factors

System Interpretation & Operational Implications

The system extended beyond individual screening to reveal structural patterns shaping student mental health.

- **Risk as a spectrum**
 - Findings showed distributed levels of vulnerability, supporting tiered intervention approaches rather than binary classification.
- **Developmental effects**
 - Higher risk in older cohorts suggested accumulating psychological strain, highlighting the importance of early intervention.
- **Interacting drivers**
 - Internal symptoms and external pressures (academic, family) emerged as interdependent factors, requiring coordinated responses beyond individual counselling.
- **From detection to system insight**
 - The system surfaced broader stress patterns (e.g. workload, environment), enabling a shift from reactive support to preventative, system-level thinking.

Research Outputs and Delivery

Self-administered online assessment platform; school-level analytical reporting and statistical analysis; stakeholder presentation of findings; individual student reports for targeted follow-up.